

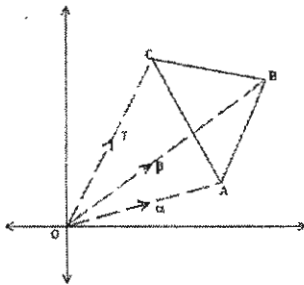
Complex Number 2

1. Let α and β be the two complex roots of $z^3 = 1$. Show that:
 - a. $\beta = \bar{\alpha}$
 - b. $\alpha^2 = \beta$ and $\beta^2 = \alpha$
 - c. $1 + \alpha + \alpha^2 = 0$
 - d. The sum of the first n terms of the series $1 + \alpha + \alpha^2 + \alpha^3 \dots$ is either 0, 1, or $-\alpha^2$, depending on the remainder when n is divided by 3.
2. Let $z = 3 + i$ and $w = 1 + 2i$. Plot the points representing each group of complex numbers on separate Argand diagrams.
 - a. $z, iz, -z, -iz$
 - b. $w, iw, -w, -iw$
 - c. $z, \bar{z}, w, \bar{w}$
 - d. $z, w, z + w$
 - e. $z, w, z - w$
 - f. $z, w, w - z$
3. Show that the points representing $-\bar{z}$ is a reflection of the point representing z in the y -axis.
4. Consider the points represented by the complex number $z, \bar{z}, -z$ and $-\bar{z}$. Show that these points form a rectangle by using:
 - a. Coordinate geometry to show that the diagonals are equal and bisect each other
 - b. The geometry of conjugates and opposites.
5. In the text it was proven that when z is complex, iz is a rotation by $\frac{\pi}{2}$ about the origin. Prove the same result when z is:
 - a. real;
 - b. imaginary.
6. The region R in the argand diagram is defined by $|z - 1| \leq |z - i|$ and $|z - 2 - 2i| \leq 1$
 - a. Sketch the region R
 - b. If z describes the boundary of the region R , find
 - i. The value of $|z|$ when $\arg z$ has the smallest value
 - ii. z in the form of $a + ib$ when $\arg(z - 1) = \frac{\pi}{4}$
7. Use de Moivre's Theorem to show that $(\cot \theta + i)^n + (\cot \theta - i)^n = \frac{2 \cos n\theta}{\sin^n \theta}$
 - a. Show that the equation $(x + i)^5 + (x - i)^5 = 0$ has roots $0, \pm \cot \frac{\pi}{10}, \pm \cot \frac{3\pi}{10}$
 - b. Hence show that the equation $x^4 - 10x^2 + 5 = 0$ has roots $\pm \cot \frac{\pi}{10}, \pm \cot \frac{3\pi}{10}$
 - c. Hence show that $\cot \frac{\pi}{10} = \sqrt{5 + 2\sqrt{5}}$
8. P is a point in the complex plane representing the complex number z , where z satisfies $|z - 2| = 2$ and $0 < \arg(z) < \frac{\pi}{2}$.
 - a. Sketch the locus described by these conditions.
 - b. Find the value of the real number k if $\arg(z - 2) = k \arg(z^2 - 2z)$
9. Let ω be the complex number satisfying $\omega^3 = 1$ and $\text{Im}(\omega) > 0$
 - a. Show that $1 + \omega + \omega^2 = 0$
 - b. Simplify $\omega^4 + \omega^6 + \omega^8$
 - c. Show that $\frac{1}{\omega^2}$ is a zero of $P(x) = x^4 + 3x^3 + 2x^2 + x - 1$

10. Let $\omega = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$

a. Show that $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$

The points A, B, C in the Argand diagram represent the complex numbers α, β, γ respectively
 ΔABC is equilateral.



b. Show that $\alpha - \gamma = \omega(\gamma - \beta)$

c. Deduce that $\alpha + \omega\beta + \omega^2\gamma = 0$

d. Explain why $\alpha, \omega\beta, \omega^2\gamma$ are the roots of a cubic equation $z^3 + pz + q = 0$

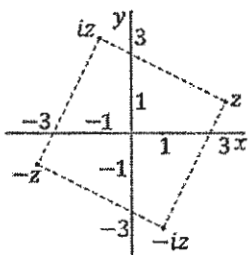
e. Deduce that $q = -\alpha\beta\gamma$

f. Prove that $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$

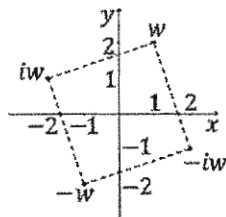
Answers:

3.

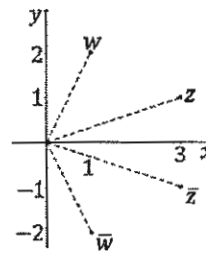
a.



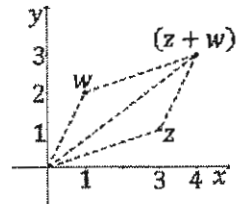
b.



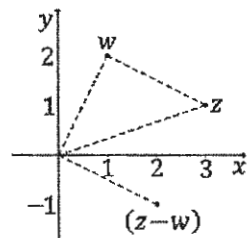
c.



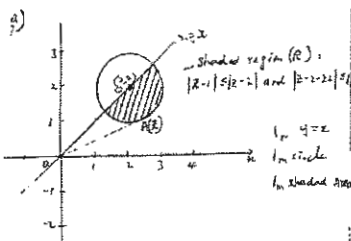
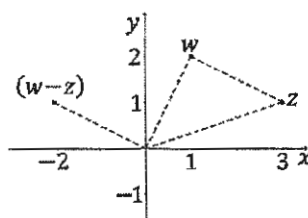
d.



e.



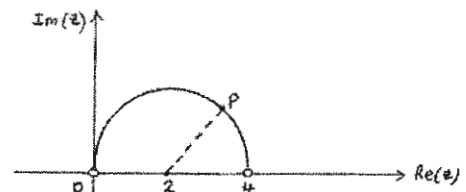
f.



6. a.

b. $|z| = \sqrt{7}$ and $z = 2 + i$ or $z = 3 + 2i$

8. a.



b. $k = \frac{2}{3}$

9. b. 0